are properties like the number of shards, compression codec, data checks on startup, and so on. If we want to change the static settings of a live index, we must close the index to reapply the settings or re-create the index with new settings.

It is always best to instantiate an index with the required static settings because applying them as an afterthought requires the index to be shut down. Having said that, there are ways to manage index upgrades (reindexing is one form of an upgrade) with zero downtime (we looked at the re-indexing feature in the last chapter—section 5.7).

- *Dynamic settings*—Dynamic settings can be applied on a live (operational) index. For example, we can change properties like the number of replicas, allowing or disallowing writes, refresh intervals, and others on indexes that are in operation.

A handful of settings fall in both camps, so having a high-level understanding of each type helps in the long run. Let's see how we can instantiate an index with some static settings.

NOTE You can consult the official documentation to learn more about the static and dynamic settings Elasticsearch supports: http://mng.bz/1qXR.

We want to create an index with these properties: three shards with five replicas per shard, the compression codec, the maximum number of script fields, and the refresh interval. To apply these configuration settings to our index, we use a `settings` object.

Listing 6.5 Creating an index with custom settings

```
PUT cars_with_custom_settings
{
  "settings":{
    "number_of_shards":3,
    "number_of_replicas":5,
    "codec": "best_compression",
    "max_script_fields":128,
    "refresh_interval": "60s"
  }
}
```

Creates an index with custom settings

The settings object encloses the required properties.

Sets the number of shards to 3

Sets the number of replicas to 5

Changes the compression from its default value

Increases the maximum number of script fields from its default of 32

Changes the refresh interval from its default of 1 second

We instruct Elasticsearch to create an index with the settings we think are essential based on our requirements. Issuing the `GET cars_with_custom_settings` command fetches the details of the index, reflecting the custom settings for shards and replicas that we set a moment ago.

As mentioned, some settings are written in stone once the index is live (static settings), and others (dynamic settings) can be changed on a running index. If we try to change any of the static properties (for example, the `number_of_shards` property) on a live index, Elasticsearch throws an exception saying that it can't update non-dynamic